

ence bandwidth is greater than the signal's bandwidth, it is frequency-selective. On the other hand, if the bandwidth of the signal is greater, then it is flat-fading. Both can be simulated using the tool.

Pathloss models. When a signal is transmitted, attenuation on the signal increases as it propagates through space. Attenuation thus produced, termed as pathloss, is a major component in the design of the link budget in a communication system.

Using the software, the designer can make use of various pathloss models like COST 207, COST 257 and ITU model for coverage prediction.

Error detection and correction codes. These are used to ensure reliable delivery of digital data sent over communication channels. Various error detection and correction schemes like Hamming, extended Golay, cyclic redundancy check codes, BCH, Reed-Solomon codes and more can be implemented using this tool.

Complementary functionalities

Ranging from mathematical functionalities like random number generation to features including fast independent component analysis (ICA) in signal processing and OFDM in communications, the library includes a lot of functionalities in order to complement the communication system design. Let us take a quick glance at some of these.

Math functions. The library consists of modules for handling basic mathematical functions such as hyperbolic, logarithmic and exponential function. IT++ also supports numerical integration and trigonometric functions. Various matrix operations such as finding the determinant, determining the inverse of a matrix or matrix decomposition can be done using the library. The module can also be used for solving linear equation systems.

Generation of random numbers, a much-demanded mathematical

The complimenting software

In order to improve performance, the tool makes use of external software. The tool can work even without these, but with its functionalities limited. Let us have a look at some of these.

Basic Linear Algebra Subprograms (BLAS). Standard building blocks required for performing basic vector and matrix operations are provided by this software.

Linear Algebra PACKage (LAPACK). LAPACK has software routines for solving systems of simultaneous linear equations, least-squares solutions of linear systems of equations, eigen value problems, and singular value problems.

Fastest Fourier Transform in the West (FFTW). This C sub-routine library can be used for computation of Discrete Fourier Transform (DFT) in single or multi dimensions. We can use FFTW for finding DFT for inputs of arbitrary size as well as for real or complex data.

You can find all these software bundled in the CD accompanying this month's EFY Plus.

functionality in communications, is achieved using Mersenne Twister generator. Integration of single-dimensional functions and the unconditional non-linear optimisation using Quasi Newton search are also provided.

Signal processing. A communications engineer has to deal with signal processing at different levels. Various filter functions and classes are defined in this library. A lot of transformations used in time-frequency domain conversions are available in this library. This includes Discrete Fourier Transform (DFT), Fast Fourier Transform (FFT), Discrete Cosine Transform (DCT) and much more.

The library can also be used for windowing and filtering operations, both in time as well as frequency domain. It can be used for evaluation of the roots of polynomials as well as for fast ICA.

Protocol simulation. Various event based simulation classes, queue classes, Transmission Control Protocol (TCP) clients and servers, packet generators, selective repeat automatic repeat request (ARQ) and others have been implemented.

Source coding. IT++ allows the user to read and save different audio

file formats. Images can be read and saved in Portable Any Map (PNM). Scalar and vector quantiser classes are provided.

What users feel

Most users who reviewed the software have been using this tool for simulation of communication systems. It works comfortably with various OSes. In comparison with other competing software like Armadillo, users feel that many signal-processing tools present in IT++ are missing in competing software. IT++ uses its own binary format, but these offer a MATLAB script to read data. "I thought IT++ had a little steeper learning curve than Armadillo but it was also quite good to use and very easy to install, at least in Ubuntu, where it comes as a package," says a user.

There are some users who are not so comfortable with this tool because of its dependencies. "Easy to install is one of my concerns since users may not be as geeky as the code writer and they may easily give up evaluating my code just because installation is not as smooth as they expected. So IT++ is ruled out by this criterion for its many dependencies," points out a user.

Fostering research in communication

Earlier research on this software was carried out in Department of Information Theory at Chalmers University of Technology, Gothenburg, Sweden. This, along with the fact that the library is coded in C++, gave the software its name.

The software has emerged over the years, finding significant applications in all areas of communication. Nowadays, the tool is even used in machine learning and for pattern recognition. Why not give it a try? ●

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